SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudocode Design

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 –Pseudocode Design**	**Assessment:**	**Assessment Tool 1 Pseudocode Design**
Weightage:	**30% of CIA**	**Total Marks:**	**100**
CO5 Statement:	Design and develop pseudocode for implementing and evaluating basic Machine Learning techniques, including Linear Regression, Logistic Regression, and Unsupervised Learning.(BL4)		
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Section / Batch:	B. Tech AIML	**Date:**	24-08-2026

Code of Conduct

I K. lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Lahari

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

6. CART

Design pseudocode for the CART classification algorithm to select the best binary split using Gini impurity and construct a classification tree.

Problem Statement

Design a program to implement the CART classification algorithm by calculating Gini impurity for possible binary splits, selecting the split with the lowest weighted Gini impurity, and constructing a classification tree.

Algorithm

1. Start.
2. Input the labelled dataset.
3. Select the input features and target class.
4. Generate possible binary splits for each feature.
5. For each split, divide the dataset into left and right groups.
6. Calculate Gini impurity for the left group.
7. Calculate Gini impurity for the right group.
8. Calculate the weighted Gini impurity of the split.
9. Select the split with the lowest weighted Gini impurity.
10. Create a decision tree node using the best split.
11. Repeat the splitting process for the child groups until the stopping condition is reached.
12. Assign the majority class to the leaf nodes.
13. Display the selected splits and classification tree.
14. Stop.

Code / Pseudocode

START

Input dataset

Dataset =

Age	Income	Purchase
25	30000	No
30	40000	No
35	60000	Yes
40	70000	Yes
45	80000	Yes
50	90000	Yes

FOR each feature

```
    Generate possible binary splits

    FOR each possible split
        Divide dataset into Left and Right groups

        Calculate Gini(Left)
        Calculate Gini(Right)

        Weighted Gini =
        (Size of Left / Total Size) * Gini(Left)
        +
        (Size of Right / Total Size) * Gini(Right)

        Store split and weighted Gini
    END FOR
END FOR

Select split having minimum weighted Gini

Create tree node using the selected split

REPEAT the same process for child nodes
UNTIL stopping condition is reached

Assign majority class to leaf nodes

Display selected split
Display classification tree

STOP
```

Gini Impurity Formula

Gini = $1 - \Sigma(p_i^2)$

where p_i is the proportion of samples belonging to class i.

Output

Best Split = Income <= 50000

Weighted Gini Impurity = 0.20

Left Node:
  Class = No

Right Node:
  Class = Yes

Classification Tree:
  Income <= 50000
    Yes → No
    No → Yes

7. Decision Tree

Design pseudocode to predict whether a customer will purchase a product using a Decision Tree based on age, income, previous purchases, and browsing history.

Problem Statement
Design a program using a Decision Tree to predict whether a customer will purchase a product based on age, income, previous purchases, and browsing history.

Algorithm

1. Start.
2. Input customer details.
3. Select age, income, previous purchases, and browsing history as features.
4. Load the trained Decision Tree model.
5. Pass the customer features to the Decision Tree.
6. Traverse the tree according to the feature conditions.
7. Reach the corresponding leaf node.
8. Predict Purchase or No Purchase.
9. Display the prediction.
10. Stop.

Code / Pseudocode

START

Input Age
Input Income
Input PreviousPurchases
Input BrowsingHistory

Create Decision Tree

Example Tree:

```
IF PreviousPurchases > 2 THEN
   Purchase = YES
ELSE
   IF BrowsingHistory = HIGH THEN
      IF Income >= 50000 THEN
         Purchase = YES
      ELSE
         Purchase = NO
      END IF
   ELSE
      Purchase = NO
   END IF
END IF
```

Display Purchase Prediction

STOP

Input

Age = 28
Income = 60000
Previous Purchases = 1
Browsing History = HIGH

Output

Age = 28
Income = 60000
Previous Purchases = 1
Browsing History = HIGH

Prediction = PURCHASE

8. Regression Analysis

Design pseudocode for a regression analysis workflow that loads a dataset, identifies independent and dependent variables, trains a regression model, and evaluates its predictions.

Problem Statement

Design a program to perform regression analysis by loading a dataset, selecting independent and dependent variables, training a regression model, predicting the target values, and evaluating the model using Mean Squared Error and R^2 score.

Algorithm

1. Start.
2. Load the dataset.
3. Handle missing values if present.
4. Identify the independent variables X.
5. Identify the dependent variable Y.
6. Split the dataset into training and testing data.
7. Initialize the regression model.
8. Train the model using training data.
9. Predict target values using testing data.
10. Calculate Mean Squared Error.
11. Calculate R^2 score.
12. Display actual and predicted values.
13. Display MSE and R^2 score.
14. Stop.

Code / Pseudocode

START

Load dataset

Dataset =

Hours	Attendance	Marks
2	70	50
3	75	55
4	80	65
5	85	72
6	90	80
7	95	88

Handle missing values

Select independent variables:
X = Hours, Attendance

Select dependent variable:
Y = Marks

Split dataset into:
Training Data
Testing Data

Initialize Regression Model

Train model using X_train and Y_train

Predicted = Model(X_test)

```
FOR each test value
   Display Actual Marks
   Display Predicted Marks
END FOR
```

MSE = $\Sigma(\text{Actual} - \text{Predicted})^2 / n$

R2 = $1 - [\Sigma(\text{Actual} - \text{Predicted})^2 /$
$\Sigma(\text{Actual} - \text{MeanActual})^2]$

Display MSE
Display R2

STOP

Output

Regression Model Trained Successfully
Actual Values:
65 72 80

Predicted Values:
64.5 73.0 79.2

Mean Squared Error = 0.43
R^2 Score = 0.98

9. Linear Regression

Design pseudocode to implement simple linear regression for predicting house price from house area using the least-squares approach.

Problem Statement
Design a program to implement simple linear regression using the least-squares approach to predict house price based on house area.

Algorithm

1. Start.
2. Input house area values `X`.
3. Input corresponding house price values `Y`.
4. Calculate mean of X and Y.
5. Calculate the slope `m`.
6. Calculate the intercept `c`.
7. Construct the linear regression equation.
8. Input the area of the house to be predicted.
9. Calculate the predicted house price using `Y = mX + c`.
10. Display slope, intercept, equation, and predicted price.
11. Stop.

Code / Pseudocode

```
START

Input Area X =
[1000, 1200, 1500, 1800, 2000]

Input Price Y =
[200, 240, 300, 360, 400]

Calculate meanX
Calculate meanY

Calculate:

m = Σ((X - meanX) * (Y - meanY))
    --------------------------------
      Σ((X - meanX)^2)
```

c = meanY - (m * meanX)

Regression Equation:
Y = mX + c
Input NewHouseArea = 1600

PredictedPrice = m * NewHouseArea + c

Display m
Display c
Display Regression Equation
Display PredictedPrice

STOP

Example Calculation

For the given data, the relationship is:

Slope (m) = 0.20
Intercept (c) = 0

Therefore,

House Price = 0.20 × Area

For a house with area `1600` sq.ft:

Predicted Price = 0.20 × 1600
= 320

Output

Slope = 0.20
Intercept = 0

Regression Equation:
House Price = 0.20 × Area

New House Area = 1600 sq.ft

Predicted House Price = 320

10. Linear Regression

Design pseudocode to implement multiple linear regression for predicting a student's final mark using attendance, internal marks, assignment marks, and study hours.

Problem Statement
Design a program to implement multiple linear regression for predicting a student's final mark using attendance percentage, internal marks, assignment marks, and study hours as independent variables.

Algorithm

1. Start.
2. Input student dataset.
3. Select attendance, internal marks, assignment marks, and study hours as independent variables.
4. Select final mark as the dependent variable.
5. Calculate the regression coefficients.
6. Calculate the intercept.
7. Construct the multiple linear regression equation.
8. Input a new student's details.
9. Predict the final mark using the regression equation.
10. Display the predicted final mark.
11. Stop.

Code / Pseudocode

START

Input Student Dataset

Attendance	Internal	Assignment	StudyHours	FinalMark
90	80	85	6	84
80	70	75	4	74
95	90	92	7	92
75	65	70	3	68
85	75	80	5	78

Select independent variables:

X1 = Attendance
X2 = Internal Marks
X3 = Assignment Marks

X4 = Study Hours

Select dependent variable:

Y = Final Mark

Initialize Multiple Linear Regression Model

Train model using:
X1, X2, X3, X4 → Y

Calculate regression coefficients:

b1, b2, b3, b4

Calculate intercept:
b0

Regression Equation:

Y = b0 + b1X1 + b2X2 + b3X3 + b4X4

Input new student:

Attendance = 88
Internal Marks = 78
Assignment Marks = 82
Study Hours = 5

Calculate:

PredictedMark =
b0 + b1(88) + b2(78) + b3(82) + b4(5)

Display PredictedMark

STOP

Output

Multiple Linear Regression Model Trained

Independent Variables:

Attendance
Internal Marks
Assignment Marks
Study Hours

Dependent Variable:
Final Mark

Regression Equation:
Final Mark = b0 + b1(Attendance)
+ b2(Internal Marks)
+ b3(Assignment Marks)
+ b4(Study Hours)

Student Details:
Attendance = 88%
Internal Marks = 78
Assignment Marks = 82
Study Hours = 5

Predicted Final Mark = 81.5